

Order Plecoptera

Stoneflies

**Stoneflies are flat**, slender-bodied insects that start life in water, developing by incomplete metamorphosis. Adults have 2 pairs of membranous wings, but are reluctant fliers. Most are short-lived, often not feeding at all. There are about 3,000 species in 19 families.

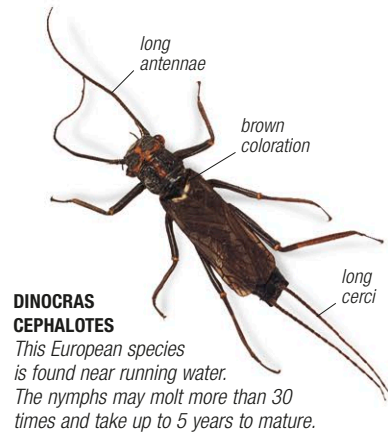
Family Perlidae

Common stoneflies

**Occurrence** 933 spp. worldwide except Australia; in lakes, rivers, vegetation near running water



As adults, common stoneflies are weak fliers, and often rest on stones close to the water's edge. They are ¾–2 in (1–4.8 cm) long, and are colored in shades of brown or yellow. The adults do not feed, but the nymphs are often carnivorous.



**DINOCRAS CEPHALOTES**  
*This European species is found near running water. The nymphs may molt more than 30 times and take up to 5 years to mature.*

Order Grylloblattodea

Rock crawlers

**Rock crawlers were first discovered** in the Canadian Rockies in 1913. Forming one family of 30 species, these small, wingless insects are adapted to cold conditions, and eat dead, windblown, or torpid prey, moss, and other plant matter. Metamorphosis is incomplete.

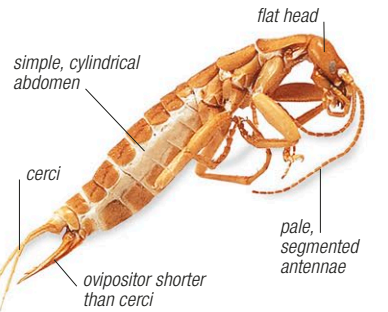
Family Grylloblattidae

Rock crawlers

**Occurrence** 30 spp. in cooler regions of the Northern Hemisphere; in limestone caves, decaying wood



Mainly nocturnal, rock crawlers, or ice crawlers, are ¾–1¼ in (1.2–3 cm) long, have small eyes, and biting mouthparts. Females may lay eggs over 2 months after mating; nymphs take years to develop.



**GRYLLOBLATTA CAMPODEIFORMIS**  
*The northern rock crawler is found at high altitudes in the USA and Canada. It lives among rocks near glaciers.*

Order Dermaptera

Earwigs

**This order contains 11 families** (about 1,900 species) of scavenging or plant-eating insects with characteristic abdominal “pincers.” Most have short front wings, and fan-shaped hindwings that can be folded. Maternal care is well developed. Metamorphosis is incomplete.

Family Forficulidae

Common earwigs

**Occurrence** 450 spp. worldwide; in leaf litter and soil, under bark or in crevices



These species are ¾–1 in (1.2–2.5 cm) long and usually dark brown. Two forcepslike appendages are straighter in females and curved in males. Females guard their eggs and lick them clean.



**FORFICULA AURICULARIA**  
*The common earwig, now found worldwide, is a pest of vegetables, cereals, fruit, and flowers.*

Order Phasmatodea

Stick and leaf insects

**These strikingly sticklike** or leaflike insects live and feed on plants, and use their remarkably effective camouflage to avoid being eaten. The males—which are rare in some species—have wings, but the females are frequently wingless. They develop by incomplete metamorphosis, and are usually nocturnal. There are about 2,500 species in 3 families.

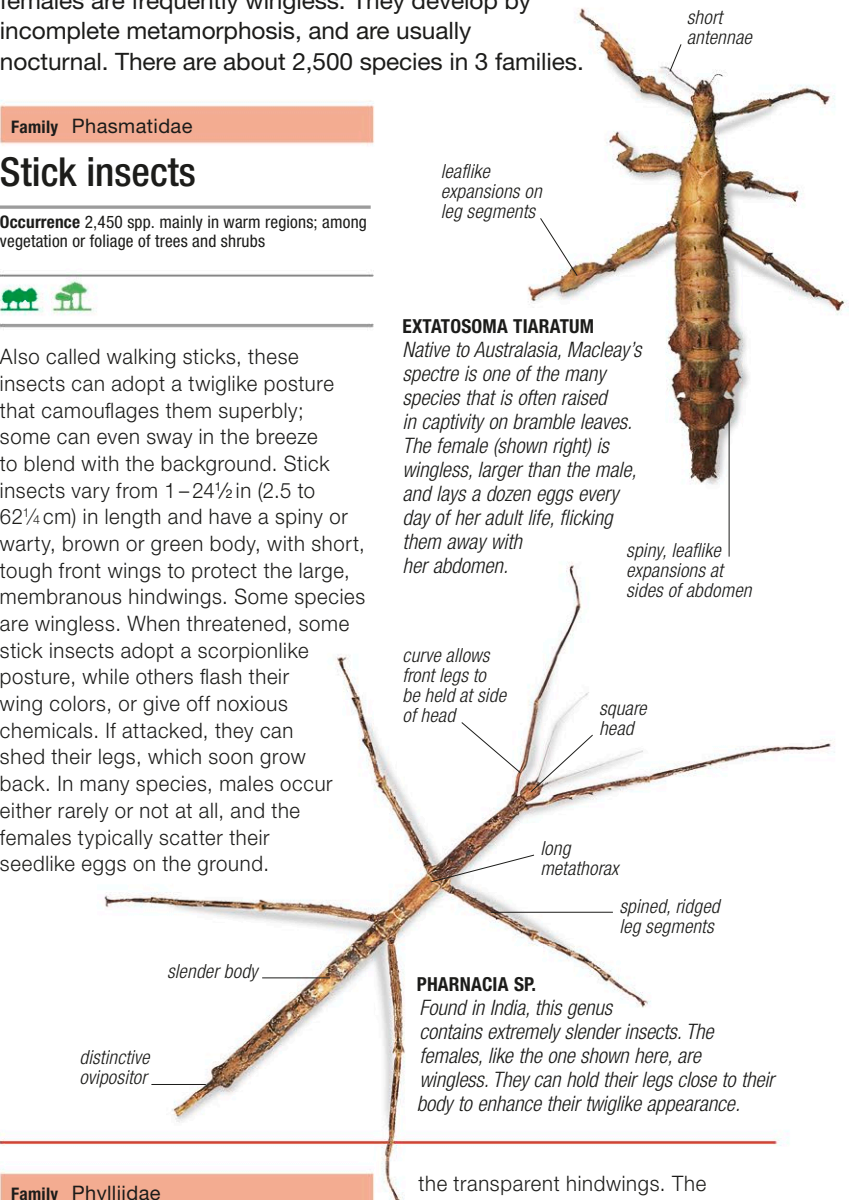
Family Phasmatidae

Stick insects

**Occurrence** 2,450 spp. mainly in warm regions; among vegetation or foliage of trees and shrubs



Also called walking sticks, these insects can adopt a twiglike posture that camouflages them superbly; some can even sway in the breeze to blend with the background. Stick insects vary from 1–24½ in (2.5 to 62¼ cm) in length and have a spiny or warty, brown or green body, with short, tough front wings to protect the large, membranous hindwings. Some species are wingless. When threatened, some stick insects adopt a scorpionlike posture, while others flash their wing colors, or give off noxious chemicals. If attacked, they can shed their legs, which soon grow back. In many species, males occur either rarely or not at all, and the females typically scatter their seedlike eggs on the ground.



**EXTATOSOMA TIARATUM**  
*Native to Australasia, Macleay's spectre is one of the many species that is often raised in captivity on bramble leaves. The female (shown right) is wingless, larger than the male, and lays a dozen eggs every day of her adult life, flicking them away with her abdomen.*

**PHARNACIA SP.**  
*Found in India, this genus contains extremely slender insects. The females, like the one shown here, are wingless. They can hold their legs close to their body to enhance their twiglike appearance.*

Family Phylliidae

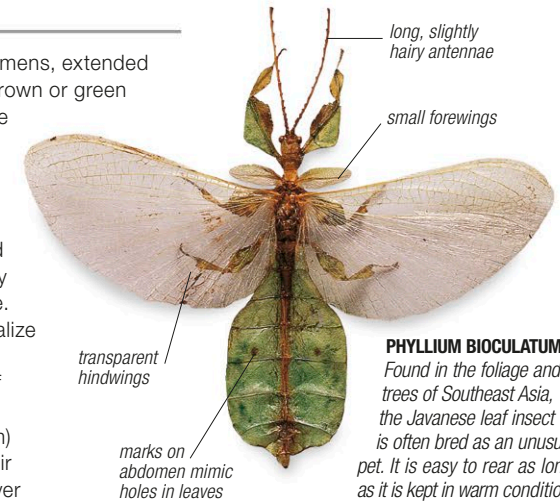
Leaf insects

**Occurrence** 30 spp. in Mauritius, Seychelles, Southeast Asia, Australasia; on various plants in well-vegetated areas



Flat, expanded abdomens, extended leg segments, and brown or green coloring all give these insects an uncanny resemblance to leaves. The disguise is helped further by blotches of color and texture, and an ability to sway in the breeze. Some species specialize in looking like dead, wrinkled leaves. Leaf insects range from 1¼–4¼ in (3 to 11 cm) in length. At rest, their veined forewings cover

the transparent hindwings. The antennae are short and smooth in females, and long and slightly hairy in males. The eggs are seedlike and are dropped on the ground. Because of their odd appearance, leaf insects are bred as pets all over the world.



**PHYLLIUM BIOCULATUM**  
*Found in the foliage and trees of Southeast Asia, the Javanese leaf insect is often bred as an unusual pet. It is easy to rear as long as it is kept in warm conditions.*

INVERTEBRATES